Facial Age and Gender Prediction using Deep Learning

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Abstract:- *In society, gender and age have two face characteristics that are highly significant. The main objectives of automated categorization are to create intelligent systems that can effectively learn and identifying faces. Deep learning methods are frequently employed in most studies and are also utilized to enhance performance. These applications use biometrics, which is generally utilized for security, as a crucial component. Emphasizing the significance of this technology, the paper discusses its potential to enhance everyday life through applications such as intelligence agencies, CCTV cameras, policing, and matrimony sites. One basic biometric technique that has gained prominence in the field of study is the face modality. The work's primary objective is intended to advance a system that recognizes the gender and age of a facial image using CNNs. We explore efficiency of custom CNN architectures versus utilizing established CNN architectures as feature extractors. Our strategy makes use of the UTK Faces dataset for gender prediction and face Age datasets for age estimation. From previous works the gender and age accuracy were 86% and 75% respectively. We obtain 89.0% accurate for gender and 78.0% accurate for age estimates by utilizing deep learning techniques. The results obtained indicate the usefulness of several pre-trained models for the task and show the value of deep learning for estimating gender and age in faces.*

Keywords: *face age detection, Convolutional neural network, Deep learning, Gender Prediction.*

I. INTRODUCTION

AI is utilized in facial age and gender detection to infer an a person's gender and age based on their visual traits. This type of technology, which can be applied to CCTV cameras, healthcare by detecting significant health hazards, may be utilized in the surveillance and security industry to instantly identify possible suspects or criminals by assessing their age and gender [2]. CNNs have proven to be exceptionally skilled in a variety of computer vision tasks., These are especially appropriate when nuanced complexities for age, gender prediction from facial images. The human face serves as a reservoir of multifaceted information, encompassing identity, emotional expression, gender, and age, all of which significantly influence non-verbal communication among individuals. Within the realm of machine learning, the prediction of gender and age from visual images stands as a pivotal task with wide-ranging applications, including access control, user-computer interaction and police enforcement. Face gender and age predictability in marketing and advertising may help to personalize marketing techniques and advertisements by evaluating the age and gender of viewers entering a business or shopping area [5]. Face recognition is used in many applications because it is considered a particularly rich input, and in recent years, it has emerged as one of the most exciting issues in pattern recognition.

Neural networks from deep learning act as the foundation for the whole model, and the neural network's neurons handle all decision-making. The objective of the article is to use the model that is being constructed to identify the parameters, such as the person's gender and age. It intends to cope with the challenge of reliably determining an individual's gender and age, which has ramifications for many industries, including marketing, healthcare, and law enforcement.

II. LITERATURE SURVEY

[1] For face recognition, deep convolutional neural networks, PCA, and SVC are used for feature extraction and classification. A Bounoua and M.K. Benkaddour provide a thorough method for face recognition that combines deep learning with more conventional machine learning approaches. Their primary the development of a facial recognition system that performs well utilizing the powers of deep convolutional neural networks (CNNs), Support Vector Machines (SVC) and Principal Component Analysis (PCA). The CNN model gains the ability to extract discriminative features—which are crucial for precise face recognition—by receiving facial picture input.

[2] Train Convolutional Neural Networks Efficiently for Face-Based Age and Gender Prediction, J. Dugelay, G. Antipov and M. Baccouche discuss how to train CNNs for reliably estimate the age and gender from face photos. These trained model's performance is assessed by the authors using suitable assessment metrics. Metrics like precision, recall, F1-score and accuracy can be applied to gender prediction. Root mean square error (RMSE) or mean absolute error (MAE) could be extensively employed metrics for age prediction.

[3] The 2014 "Age and Gender Estimation of Unfiltered Faces" paper published in IEEE Transactions on Information Forensics and Security, by Hassner, Enbar and Eidinger challenges the difficult issue with estimation of gender, age for unfiltered facial photos. Their primary goal is to create a precise method that can be used to figure out the age and gender of an individual from unprocessed face photos. To figure out how to translate the collected characteristics to the target variables, the authors use support vector machines (SVMs), random forests, and deep neural networks are examples of machine learning techniques.

[4] Age-group classification from facial images is the focus of a paper by K. Ueki, T. Hayashida, and T. Kobayashi titled Age-group classification based on subspace and using facial photos in different lighting conditions which was shown to the Proceedings of Computer Identification of Faces and Gestures. The paper specifically addresses the difficulties presented by different lighting conditions. Principal component analysis (PCA) and eigenfaces are two examples of techniques used to extract facial traits that are invariant to fluctuations in lighting. The authors use the features they have collected to create a classification model that divides people into several age groups. This model uses methods like linear discriminant analysis and support vector machines to extract discriminative age-related features from the face photos.

[5] The 2015 IEEE Workshops on Pattern Recognition and Computer Vision presentation by T. Hassner & G. Levi, gender and age Classification Using CNN discusses the use for CNNs to concurrently classify gender, age from face pictures. They create a CNN architecture that works well for the purpose of classifying the people's gender, age from facial photos. To extract hierarchical information from the input images, model probably consists of several layers of convolution, which are pooling layers. Using the proper evaluation measures, they assess the efficiency of their approach for classifying people by gender and age.

III. METHODOLOGY

Dataset Preparation

In this study, we utilized the UTKFace dataset, comprising more than 23,000 trimmed, centered face pictures including gender, ethnicity and age. With a total of 23,708 images, we encountered only six missing age labels. Notably, this dataset offers extensive variability in facial attributes, including expression, illumination, pose, resolution, and occlusion, making it ideal for our research objectives. We selected this dataset due to its relatively uniform distributions across age, gender, and ethnicity, as well as its representation of diverse characteristics such as brightness, occlusion, and positioning, reflecting the general people [3]. This diversity was deemed essential for accurately developing and assessing our models. Each image in the dataset is associated with a three-part tuple comprising the individual's age where female is represented as '1' and male is represented as '0'. To ensure standardized results, we adopted a consistent approach for both custom convolutional neural network employing the dataset partitioning strategy supporting verification, training purposes and evaluation [8]. It provides insights into the composition of the data splits concerning gender and age, accordingly.

Fig. 1. A few pictures from the UTKFace collection.

Data Preprocessing

In preparation for age and gender estimation tasks, a systematic preprocessing pipeline was implemented for image data. Firstly, images underwent resizing to standardized dimensions: 224 x 224 pixels for age estimation and 64 x 64 pixels for gender estimation, ensuring uniformity across the dataset [11]. By making sure that every image has the same size, resizing helps to avoid differences in image dimensions that might lead to differences in feature extraction and model performance. Color conversion steps varied according to task requirements: grayscale conversion for age estimation simplified processing, while RGB format was retained for gender estimation to preserve color information. Additionally, normalizing procedures were applied to bring the pixel values of the pictures into uniformity [13]. Typically, this involved scaling pixel intensities to a predetermined range (e.g., [0, 1] or [-1, 1]). Thus, pixel values were adjusted to a normalized range of 0 to 1, ensuring consistency in processing methods. By guaranteeing that input characteristics have comparable magnitudes and avoiding the dominance of certain features during model training, normalization aids in training process stabilization. Lastly, subsets of the dataset for validation as well as training were carefully formed. post-preprocessing, facilitating effective model training and evaluation. This meticulous preprocessing strategy laid a solid foundation for robust age and gender estimation model development.

Model Architecture

CNN architecture utilized for the categorization of images and recognition tasks are structured with meticulous design considerations. Comprising four CNN layers and an average pooling layer in each pair, CNN adeptly extracts hierarchical features from input images. Filter configuration is optimized, with an initial convolutional layer housing 32 filters, progressively doubling in subsequent layers to enhance the model's capacity for capturing intricate patterns. Rectified Linear Unit (ReLU) activation functions empower non-linear transformations and feature extraction across every convolutional layers [9]. A tier of global average pooling follows the fourth convolutional layer diminishes spatial dimensions, consolidating extracted features. Subsequently, the feature representation is augmented through 132 nodes make up the completely connected layer, and a softmax activation function comes next on ensuring normalized probabilities across age classes [10]. The output layer, with seven nodes corresponding to distinct age classes, facilitates precise age range prediction, thereby underscoring the CNN's efficacy in age classification tasks.

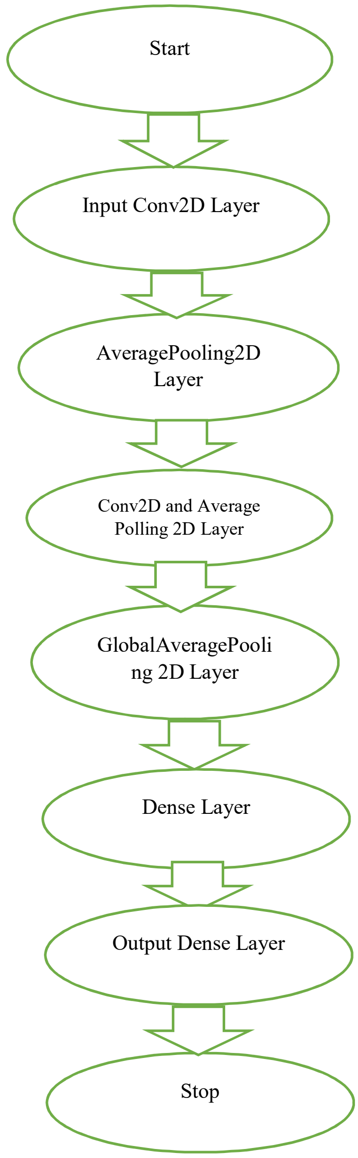

Fig. 2. Building Architecture

Model Training

The training procedure was carried out in an effort to create reliable models for Convolutional Neural Networks (CNNs)-based face age and gender detection [15]. To enable thorough model assessment, the preprocessed UTKFace dataset which was enhanced with annotations for age and gender attributes were first split into separate subsets for verification, evaluation, and training. The model acquired to distinguish between different face features during training, identifying complex patterns that correspond to attributes of gender and age.

- The Adam optimizer, renowned for its effectiveness in optimizing deep neural networks, was used to train the model with a fixed learning rate of 0.001.
- Training data was divided into batches of size 32, enhancing computational efficiency and facilitating parallel processing during model training.
- The model underwent training for a total of 30 epochs, representing how many times the neural network processed the whole training dataset both forward and backward.
- Categorical loss function used for this was cross-entropy for age estimation, measuring the dissimilarity between predicted and actual age class distributions [2].
- As the loss function, binary cross-entropy was utilized for gender estimation, quantifying the disparity between predicted gender probabilities and ground truth labels.

Design Assessment

- The newly developed model was tested for both gender and age estimation tasks on the verification group., ensuring unbiased assessment of its performance.
- Accuracy and Mean Absolute Error (MAE) were utilized as evaluation metrics for age estimation, quantifying both classification accuracy additionally the mean absolute difference between expected and measured ages.
- Accuracy, F1-score were employed as evaluation metrics for gender estimation, providing insights into how well the model can categorize gender labels and balance between precision and recall [6].
- The evaluation results were juxtaposed with previous works in the field, enabling a thorough evaluation of the suggested model's capabilities against existing methodologies.
- The model attained accuracy rates around 78% for estimating age and 89% for estimating gender on the validation set, showcasing its efficacy and robustness in successfully guessing age and gender based on face.
- Total obtained results underscored efficacy to CNN-based method for estimating gender and age from facial expressions, highlighting its potential for real-world applications in various domains.

Age Detection Model

- Input Layer: Conv2D with 32 filters, applies convolution operation to input images.
- After the input layer, the AveragePooling2D Layer applies average pooling to feature maps in order to minimize their spatial dimensions.
- Three Pairs of AveragePooling2D and Conv2D Layers: The filters in each pair have been adjusted gradually to 256, 128 and 64, correspondingly, to gather additional intricate details.
- It calculates every attribute map's average across all spatial areas, producing one value.
- Complete dense layered network of 132 units, introduces non-linearity and learns complex patterns.
- The last dense stage uses the softmax activation method to produce a stochastic result.

Gender Prediction Model

- For preserving most noticeable features, the model employs MaxPooling2D layers rather than AveragePooling2D layers after each Conv2D layer. It does this by choosing the maximum value from each pooling window.
- The final dense layer uses the function of soft max activation to produce probabilistic output and comprises three nodes, each of which represents one of the three gender labels (male, female, or other) in the collection.

IV. TRAINING AND EVALUATION

Implementing the architecture for a CNN, gender, age detection models were prepared 75% of the information and examined on 25% of data [5]. Three pairs of Conv2D layers (one with 64 filters, the other with 128 filters, and so on) together with AveragePooling2D layers, a 132.0 units in the Dense layer, seven units in the final Dense layer, and the GlobalAveragePooling2D layer were used to develop the age detection model. A Dense level with 128.0 nodes, an final Dense level with 3 units, 3 sets of Conv2D layers, each containing 256, 128, and 64 filters, respectively, and the gender recognition architecture consisted of a MaxPooling2D structure and a source Conv2D structure with 32 effects. While the Grayscale pictures in the Dataset were used for training the gender detection algorithm, and 234,000 enhanced. Based on validation accuracy, a Model Checkpoint callback object was built to preserve the best model. With a batch size of 512 and 60 epochs of training, the model's shuffle parameter was set to False inorder to improve repeatability. During this procedure, its Tensor Board push out had been utilized for illustrating the learning progress. A line chart was used to plot the accuracy and decline values To be able to assess the efficiency of the model [12]. The chart revealed an enhancement regarding accuracy values, reducing loss values during course of epochs. The final structure is utilized for estimating the gender and age of faces in fresh photos, with 78.0% and 89.0% accuracy rates for gender and age identification, respectively.

Fig. 3. Sample Images from UTK Faces

V. RESULT

Two measures are available to assess the effectiveness of the suggested methodology: classification accuracy and loss rate. By splitting the entire amount of photographs in a subset by the amount of correctly identified photos in that subset, the rate of identification is computed. For every subset of data, including age and gender classification, this metric can be computed. The categorization score for age and gender are added together yield the overall accuracy rate [14]. On the other hand, the reduction rates for gender and age categories add up to total decline rate. Difference between expected and actual values is measured by the loss rate; a smaller number denotes greater performance. All things considered, these metrics allow us to assess how well the suggested methodology performs in correctly identifying photographs based on gender and age.

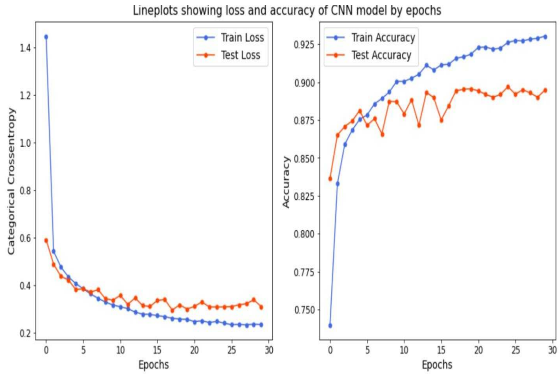

Fig. 3. CNN model's accuracy and loss for the gender

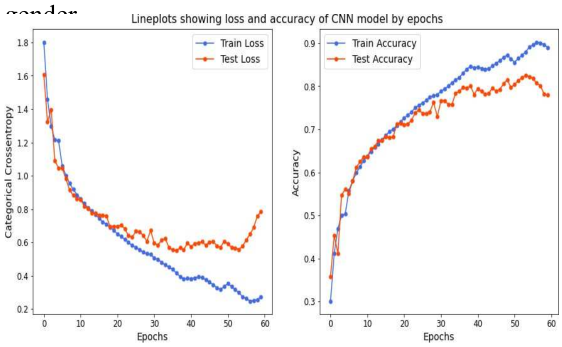

Fig. 4. CNN model's loss and accuracy for the age

VI. CONCLUSION

Our CNN model for age prediction was built using an initial Conv2D layer with 32 filters, and it was trained on a dataset of 234,000 enhanced grayscale face photos. The GlobalAveragePooling2D layer was the result of three sets for Conv2D extends AveragePooling2D structure with 256, 128 and 64 effects, respectively, following this. First Dense level with 132.0 units and a final Dense level with seven units, representing various age spans , were combined [10]. The model performed well, achieving an accuracy on the test dataset that was higher than 83.5%. However, it's critical to identify any potential differences in performance between circumstances and populations. Despite this, the age prediction CNN model holds promising prospects across various domains including law enforcement, healthcare, and marketing. Continuous refinement and research endeavors stand to enhance the model's accuracy and applicability across diverse settings.

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